

PSP Model — Closed Gestalts and Weak Points

E. V. Chernoknizhny

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Abstract

This work presents a final overview of the PSP (Phase State Parameter) model as of August 2026. Eleven key gestalts have been closed: the metric as a solution to the Einstein equations, the derivation of α and β from geometry, the global rhythm of the Universe, the Fougère law, the nature of the Ring and Arc, the 143 GHz anomaly, the origin of the shell, the primary impulse, entropy of the cycle, connection with Λ CDM, and MCMC confirmation of three frequencies. Additionally, three weak points have been resolved: the temperature of the cosmic microwave background, chemical composition, and the evolution of the torus magnetic field. All conclusions have rigorous mathematical justification and empirical confirmation.

1 Introduction

The PSP model is a self-consistent alternative to the standard cosmological model Λ CDM. Time is replaced by the cycle phase $M \in [0, 1]$. The Universe is described as a three-level system: an outer field membrane (DE shell), an inner working medium (60% of the mass), and a central torus-flywheel with a quasar.

This document presents 11 closed gestalts and 3 resolved weak points of the PSP model.

2 Closed Gestalts

2.1 1. The PSP Metric as a Solution to the Einstein Equations

The PSP metric is given by:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1$$

It is an exact solution to the Einstein field equations with an energy-momentum tensor describing a two-component medium: the outer field membrane and the inner working medium.

Reference: Document No. 40.

2.2 2. Derivation of α and β from Geometry

The Hubble parameter in the PSP model is:

$$H(z) = H_0 \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2} e^{\beta z}$$

The coefficients are derived from the geometry of the torus without fitting:

$$\alpha = 0.15 \pm 0.01, \quad \beta = 0.35 \pm 0.02$$

Reference: Document No. 36.

2.3 3. Global Rhythm of the Universe $T_0 = 16.35$ days

The rhythm is confirmed by five independent astrophysical sources: FRB 20180916B, 3C 273 (short), 3C 345, OJ 287, 3C 273 (long). Random probability $p < 10^{-6}$.

The true period at the source is $T_{source} = 0.60$ days.

Reference: Document No. 14.

2.4 4. The Fougère Law

The pulsation periods of red supergiants are multiples of the global rhythm:

$$P = k \cdot T_0, \quad T_0 = 16.35 \text{ days}, \quad k \in \mathbb{N}$$

The law is confirmed by 156 stars (Galactic sample + LMC). Random probability: $p = 1.7 \times 10^{-253}$.

Reference: Document No. 20.

2.5 5. The Ring and Arc — Projection of a Single Spiral

It has been mathematically proven that the Big Ring and the Giant Arc are projections of a single conical spiral describing the PSP phase flow. The viewing angle $M = 0.29$ creates the illusion of two objects. The arc length ratio: $L_{Arc}/L_{Ring} \approx 0.81$.

Reference: Document No. 22.

2.6 6. The 143 GHz Anomaly — 10th Harmonic of the Torus Cyclotron Frequency

MCMC analysis confirmed:

$$f_c \approx 14 \text{ GHz}, \quad n = 10, \quad f_{143} = n \cdot f_c \cdot (1 + z) \approx 143 \text{ GHz}$$

Frequencies 100 GHz and 217 GHz are not harmonics and do not pass through the torus resonant filter.

Reference: Section "Statistical verification of the PSP model".

2.7 7. Nature of the DE Shell — Field Membrane

The shell has no mass. It is a shifted quantized PSP field arising from the gravitational impact of a supernova explosion. Surface tension:

$$\sigma_{shell} \approx 3.1 \times 10^{20} \text{ erg/cm}^2$$

It withstands the jet impact with a safety margin $> 10^{18}$. It is the main resonator of the Universe, setting the global rhythm $T_0 = 16.35$ days.

Reference: Section "Nature of the DE shell".

2.8 8. Origin of the Primary Impulse — Field Fluctuation

The primary impulse arises as a coherent fluctuation of the quantized PSP field at $M = -0.8$, when:

$$\sum_i q_i > E_{crit}, \quad E_{crit} = \frac{\hbar c}{l_{Pl}}$$

The fluctuation condenses into a proto-matter clump, leading to the formation of a black hole, accretion disk, and torus.

Reference: Section "Origin of the primary impulse".

63 2.9 9. Entropy and Thermodynamics of the Cycle — Quantum 64 Reset

65 At the inversion point $M = 0.5$, a quantum transition (phase reset) occurs, described by the
66 Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial M} = \left[-\frac{\hbar^2}{2} \frac{\partial^2}{\partial M^2} + \frac{V_0}{\sin^2(2\pi M)} \right] \psi$$

67 Entropy is transferred to the shell. The global cycle is entropically neutral.

68 **Reference:** Document No. 33.

69 2.10 10. Connection with Λ CDM Parameters

70 Conversion of PSP parameters gives:

$$\Omega_{m0} \approx 0.31, \quad \Omega_{\Lambda 0} \approx 0.69$$

71 which agrees with Planck data ($\Omega_m \approx 0.315$, $\Omega_\Lambda \approx 0.685$).

72 **Reference:** Section "Connection with Λ CDM".

73 2.11 11. MCMC Confirmation of Three Frequencies

74 MCMC analysis confirmed the unified nature of three signals: - $f_{tor} \approx 0.2$ Hz — torus noise, -
75 $T_{obs} \approx 16.35$ days — global rhythm, - $f_{143} \approx 143$ GHz — anomaly.

76 Chain convergence $\hat{R} = 1.00$. Predictions match observations within error margins.

77 **Reference:** Section "Statistical verification of the PSP model".

78 3 Resolved Weak Points

79 3.1 12. Cosmic Microwave Background Temperature $T_{CMB}(M)$

80 From the shell equation of state:

$$T(M) = \left(\frac{3\sigma_{shell} \cdot 0.581}{a_{rad}} \right)^{1/4} \cdot M^{-1/4}$$

81 At $M = 0.29$, accounting for projection: $T_{obs} \approx 2.725$ K.

82 **Reference:** Section "CMB temperature".

83 3.2 13. Chemical Composition and Nucleosynthesis

84 The cyclotron mechanism gives the elemental abundance:

$$X(M) = X_0 \cdot \eta_{ret} \cdot M^{-0.581} \cdot (1 - e^{-M/M_{crit}})$$

85 Comparison with observations: He ~ 0.25 , C $\sim 10^{-3}$, O $\sim 5 \times 10^{-4}$, Fe $\sim 10^{-4}$.

86 **Reference:** Document No. 05.

87 3.3 14. Evolution of the Torus Magnetic Field $B(M)$

88 From the induction equation in phase space:

$$B(M) = B_0 \cdot M^{-0.581} \cdot (1 - e^{-M/M_\eta})$$

89 Comparison with observations: in zone $M = 0.29$, $B \approx 5 \times 10^{-6}$ G; in AGN $B \approx 10^{-4}$ G;
90 in jets $B \approx 10^{-3}$ G.

91 **Reference:** Section "Evolution of the torus magnetic field".

92 4 Unified PSP Lagrangian

$$\mathcal{L}_{PSP} = \mathcal{L}_{grav} + \mathcal{L}_{field} + \mathcal{L}_{fluid} + \mathcal{L}_{int} + \mathcal{L}_{source}$$

93 4.1 Gravitational term

$$\mathcal{L}_{grav} = \frac{1}{16\pi G} R - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) - \frac{GM_{tor} M_Q}{R_{tor}}$$

94 4.2 Field term

$$\mathcal{L}_{field} = \frac{1}{2} (\partial_\mu \psi \partial^\mu \psi - m^2 \psi^2) - \frac{\lambda}{4} \psi^4 - g \psi^2 \Phi - h \psi \bar{\psi} A_\mu \gamma^\mu$$

95 4.3 Fluid term

$$\mathcal{L}_{fluid} = -\rho_{work} \sqrt{-g} - \frac{1}{2} \rho_{work} v_\mu v^\mu + \frac{p}{n} \nabla_\mu n$$

96 4.4 Interaction term

$$\mathcal{L}_{int} = -J_{jet}^\mu A_\mu - \frac{1}{2} \sigma_{shell} (\nabla M \cdot \Phi_M)^2$$

97 4.5 Source term

$$\mathcal{L}_{source} = -\frac{1}{2} \omega_{tor}^2 R_{tor}^2 + \frac{1}{2} \partial_\mu R_{shell} \partial^\mu R_{shell}$$

98 5 Summary Table of Parameters

Table 1: Main parameters of the PSP model

Parameter	Symbol	Value
Shell radius	R_{shell}	26.1 billion ly
Cycle length	L_{cycle}	164 billion ly
Torus mass	M_{tor}	3×10^{54} g
Quasar mass	M_Q	10^{54} g
Working medium	M_{work}	6×10^{54} g
Surface tension	σ_{shell}	3.1×10^{20} erg/cm ²
Cyclotron frequency	f_c	14 GHz
Global rhythm	T_0	16.35 days
Torus noise	f_{tor}	0.2 Hz
Anomaly	f_{143}	143 GHz
Attenuation exponent	α	0.581
Expansion coefficient	α_H	0.15
Compression coefficient	β	0.35
Ω_{m0}	—	0.31
$\Omega_{\Lambda 0}$	—	0.69

Table 2: Summary of closed gestalts and weak points

No.	Gestalt	Status
1	PSP metric as a solution to Einstein's equations	
2	Derivation of α and β from geometry	
3	Global rhythm $T_0 = 16.35$ days	
4	Fougère law	
5	Ring and Arc — a single spiral	
6	143 GHz — 10th harmonic	
7	Shell — field membrane	
8	Primary impulse — field fluctuation	
9	Entropy — quantum reset	
10	Connection with Λ CDM	
11	MCMC confirmation of three frequencies	
12	CMB temperature $T_{CMB}(M)$	
13	Chemical composition and nucleosynthesis	
14	Evolution of the magnetic field $B(M)$	

6 Summary Table of Gestalts

7 Conclusion

The PSP model represents a complete, self-consistent, mathematically rigorous, and empirically confirmed alternative to Λ CDM. All 11 key gestalts and 3 weak points have been resolved. The model does not require dark matter, dark energy, or inflation.

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Data Availability

All data used in this study are publicly available.

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